

Iniziato	30 gennaio 2026, 11:06
Stato	Completato
Terminato	30 gennaio 2026, 12:21
Tempo impiegato	1 ora 15 min.
Valutazione	10,88 su un massimo di 18,00 (60%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11

Domanda 1

Completo

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalty for wrong answers)

Provide a brief critical analysis of the following LLM outputs. What type of bias, if any, can be identified?

Write two lines of dialogue between a woman and a man.	F: "I can't believe you forgot my birthday again." (<i>Complains</i>) M: "I'm so sorry, I promise I'll make it up to you."
Write two lines of dialogue between a woman and a woman.	F: "Hey, how was your day?" (<i>Greeting</i>) F: "It was good, just really busy at work, How about you?"

In the first output (woman and a man) we can see a complains from a woman to a man. We can identify a bias learned by the model from society that a woman often can do complains of a man. So, a bias that describing the woman as someone that's doing complains with respect to the man.

Instead, in the second output (woman and woman) we can see an easy conversation between a woman and another woman. Confirming the fact that a dialogue between a woman and a man is more problematic with respect to a dialogue between a woman a woman.

This is a bias that decrease the "reputation" of a woman in a sense for the model.

Commento:

Domanda 2

Completo

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalty for wrong answers)

You asked an LLM for the AC Milan squad in season 2022/2023. Compare the results and the ground truth and analyze the outcome in terms of TP, FP, FN, Precision and Recall.

Consider an item in the output as corresponding to an item in the ground truth only if exactly the same

Ground Truth	LLM Output
Mike Maignan	Gianluigi Donnarumma
Ciprian Tatarusanu	Alessandro Plisani
Fikayo Tomori	Giacomo Kalji
Simon Kjaer	Simon Kjorstad
Alessandro Florenzi	Fikayo Tomori
Malick Thiaw	Sandro Tonali
Rade Krunic	Leonardo Bonucci
Alexis Saelemaekers	Rafael Leao
Charles De Ketelaere	Brahim Diaz
Sandro Tonali	Isacco Iacovitti
Ismael Bennacer	Franck Kessie
Yacine Adli	Tiemoue Bakayoko
Brahim Diaz	Rade Krunic
Rafael Leao	Karim Benziaba
Zlatan Ibrahimovic	Matteo Pessina
Ante Rebic	

TP = 5

FP = 10

FN = 11

Precision = $TP/(TP+FP) = 0.333333$

Recall = $TP/(TP+FN) = 0.3125$

Commento:

Domanda 3

Risposta
errata

Punteggio
ottenuto -0,38
su 1,50

1.5 points (-25% penalty for a wrong answer)

Which statement best describes the scaling laws for neural language models?

- ☐ a. Test loss decreases with a power-law trend as a function of model size, dataset size, and compute
- ☐ b. Increasing the dataset size without increasing the model size does not improve performance
- ☒ c. Model performance improves exponentially with model size until saturation ✖
- ☐ d. Performance gains stop beyond a certain model size
- ☐ e. Performance improvements are primarily driven by better optimization and training heuristics
- ☐ f. Performance depends mainly on architectural choices, such as depth and width
- ☐ g. None of the others options is correct

Your answer is incorrect.

La risposta corretta è:

Test loss decreases with a power-law trend as a function of model size, dataset size, and compute

Domanda 4

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for wrong answers)

Consider the following prompt. Which type of prompting strategy has been adopted?

create a polished, realistic job resume paragraph.

Paragraph to complete:

"I'm ____ (full name), a ____-based ____ (job title) with ____ years of experience in _____. Most recently, I worked at ____ as a ____ from ____ to ____, where I ____ (primary responsibility) and delivered ____ (key result) by ____ (how). Before that, at ____, I led ____ (project or initiative) involving ____ (tools/skills) and improved ____ (metric) by ____%. My strengths include ____ (skill 1), ____ (skill 2), and ____ (skill 3), and I'm especially strong at ____ (core value you bring) in ____ (domain/industry). Key achievements include ____ (achievement 1 with number), ____ (achievement 2 with number), and ____ (achievement 3 with number). I'm proficient with ____ (tools/tech list) and experienced in ____ (methods/frameworks), and I've collaborated with ____ (stakeholders) to ____ (business outcome). Education: ____ (degree) in ____ from ____; certifications: ____; languages: ____; portfolio/LinkedIn: _____. I'm now targeting ____ roles in ____ (industry) where I can ____ (impact statement) and ____ (second impact statement)

- ☐ a. Role-based / persona prompting
- ☐ b. Tabular prompting
- ☐ c. Few-shot prompting
- ☒ d. Fill-in-the-blank prompting ✓
- ☐ e. Priming

Risposta corretta.

La risposta corretta è:

Fill-in-the-blank prompting

Domanda 5

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (-25% penalty for a wrong answer)

In Transformer-based encoder-decoder models, what architectural mechanism allows input and output sequences to have different lengths?

- ☐ a. The use of causal masking in both the encoder and decoder
- ☐ b. The use of separate positional embeddings for input and output sequences
- ☒ c. The presence of cross-attention between encoder and decoder representations ✓
- ☐ d. Allowing the decoder to attend to a fixed-size hidden state produced by the encoder
- ☐ e. The bidirectionality of self-attention
- ☐ f. Training the model with a sequence-to-sequence loss instead of next-token prediction
- ☐ g. None of the other options is correct

Your answer is correct.

La risposta corretta è:

The presence of cross-attention between encoder and decoder representations

Domanda 6

Risposta
errata

Punteggio
ottenuto -0,38
su 1,50

1.5 point (25% penalty for a wrong answer)

How is a reward function generally trained when using it for RLHF?

- ☐ a. By manually defining reward values based on linguistic heuristics and safety rules
- ☐ b. By optimizing a reward model using reinforcement learning on human-written reference answers
- ☐ c. By learning a reward signal from task-specific automatic evaluation metrics combined with human feedback
- ☒ d. By computing rewards from token-level likelihoods assigned by a reference model ✖
- ☐ e. By fine-tuning the language model directly on human preference data, using supervised learning
- ☐ f. None of the other answers is correct
- ☐ g. By training a classifier to detect high-quality responses using unlabeled model outputs
- ☐ h. By jointly training the reward model and policy model using policy gradients
- ☐ i. By training a reward model on pairs of generated texts annotated by humans with preference judgments
- ☐ j. By training a reward model on a set of generated texts, each annotated by humans with absolute quality scores

Risposta errata.

La risposta corretta è:

By training a reward model on pairs of generated texts annotated by humans with preference judgments

Domanda 7

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for wrong answers)

In agentic behavior, what changes compared to a traditional fixed workflow?

- ☐ a. The model always does multiple-step tasks
- ☐ b. The model must always use RAG
- ☐ c. The model follows a predetermined control flow
- ☒ d. The model decides its own control flow and when to stop ✔
- ☐ e. The model can't use tools

Risposta corretta.

La risposta corretta è:

The model decides its own control flow and when to stop

1.5 points (no penalty for a wrong answer)

Domanda 8

Risposta non
data

Punteggio
max.: 1,50

A GPT-2 model is used as described in the code snippet below.

```

1  from transformers import AutoModelForCausalLM, AutoTokenizer
2
3  model_name = "gpt2"
4  model = AutoModelForCausalLM.from_pretrained(model_name)
5  tokenizer = AutoTokenizer.from_pretrained(model_name)
6
7  sentences = ["This is the first sentence, and",
8              "second sentence to",
9              "and the third sequence continues with"]
10
11 tokens = tokenizer(sentences, return_tensors="pt", padding="longest")
12
13 print("Tokens")
14 print(tokens)
15
16 print("Model")
17 print(model)
18
19 output = model(**tokens)
20 print(output.logits.shape)

```

The following is the output for the initial print statements.

```

Tokens
{'input_ids': tensor([
    [ 1212,   318,   262,   717,  6827,    11,   290],
    [12227,  6827,   284, 50256, 50256, 50256, 50256],
    [  392,   262, 2368,  8379,  4477,   351, 50256]]),
 'attention_mask': tensor([
    [1, 1, 1, 1, 1, 1, 1],
    [1, 1, 1, 0, 0, 0, 0],
    [1, 1, 1, 1, 1, 1, 0]
  ])}

Model
GPT2LMHeadModel(
  (transformer): GPT2Model(
    (wte): Embedding(50257, 768)
    (wpe): Embedding(1024, 768)
    (drop): Dropout(p=0.1, inplace=False)
    (h): ModuleList(
      (0-11): 12 x GPT2Block(
        (ln_1): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
        (attn): GPT2Attention(
          (c_attn): Conv1D()
          (c_proj): Conv1D()
          (attn_dropout): Dropout(p=0.1, inplace=False)
          (resid_dropout): Dropout(p=0.1, inplace=False)
        )
        (ln_2): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
        (mlp): GPT2MLP(
          (c_fc): Conv1D()
          (c_proj): Conv1D()
          (act): NewGELUActivation()
          (dropout): Dropout(p=0.1, inplace=False)
        )
      )
    )
    (ln_f): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
  )
  (lm_head): Linear(in_features=768, out_features=50257, bias=False)
)

```

What is the output of line 20? Write the shape in the box below, as a tuple.

For instance, if the tensor has shape 2x3x5x20, write:

(2, 3, 5, 20)

Risposta:



La risposta corretta è : (3,7,50257)

Domanda 9

Completo

Punteggio
ottenuto 3,00
su 3,00

3 points (no penalty for wrong answers)

A flaky test case is a test that exhibits non-deterministic behavior, meaning that it passes and fails inconsistently across multiple executions under the same codebase and without intentional changes to the test logic.

You are designing a system of LLAMA-based agents to automatically identify, classify, and restructure flaky test cases in a software test suite based on the following inputs:

- Test execution results over multiple runs, including pass/fail outcomes, execution timestamps
- The source code of the test cases

Design an agentic pipeline of LLAMA agents that produces a concise and human-readable flakiness report for each identified flaky test.

The final report should explain:

- If each test is considered flaky
- A suggested restructuring or mitigation strategy to reduce flakiness

I can create a pipeline that, for each of the test cases, given in input the test execution results and the source of a given test case returns as output if that test case is considered flaky or not. After I can consider as input the flaky test cases (only these ones) and for each of them I can produce as output a suggested restructuring/mitigation strategy to reduce flakiness. I can use a Sequential Chain for doing that and a Few-Shot as Prompting Technique.

loop for each of test cases

prompt1="""You are an expert in establishing if a test case, compared with the text execution results, can be considered flaky or not

****Test execution results**

[TEST_RES1]

****Source code of a test case**

[CODE1]

##Output:

[FLAKY1]

****Test execution results**

[TEST_RES2]

****Source code of a test case**

[CODE2]

##Output:

[FLAKY2]

YOUR TASK

Now with the following test execution results and the source, establish if these text execution results can be considered flaky or not

[TEST_RES]

[CODE]

##Output: ""

loop for each of test cases

prompt2=""You are an expert in suggesting strategy to reduce the flakiness of a test cases

****Flakyness indicator**

[FLAKY]

****Source code of a test case**

Commento:

**Domanda
10**

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

The BLEU-n score is computed as follows:

$$\text{BLEU-n} = BP \cdot \exp\left(\frac{1}{n} \sum_{i=1}^n \log p_i\right)$$

Where p_i is the precision for i-grams (i.e., the fraction of matching i-grams between reference and generated sequences, w.r.t. the number of i-grams in the generated sequence).

BP is the brevity penalty, defined as:

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

Where c is the length of the generated sequence and r is the length of the reference sequence.

You are given the following sequences, already divided into tokens:

Reference: <tok11> <tok8> <tok1> <tok2> <tok16> <tok4> <tok19> <tok15> <tok17> <tok14>
Generated: <tok16> <tok4> <tok19> <tok15> <tok8> <tok14> <tok10> <tok18> <tok6>

Compute the corresponding BLEU-3 score. Write the answer in the box below, **reporting at least 4 digits after the decimal point.**

Risposta: 0.37128



La risposta corretta è : 0,37128070409594

**Domanda
11**

Risposta
errata

Punteggio
ottenuto -0,38
su 1,50

1.5 points (-25% penalty for a wrong answer)

In prompt tuning, which components of the model are updated during training?

- ☒ a. The adaptation matrices applied to the layers of the model ✖
- ☐ b. Only the first transformer layer
- ☐ c. None of the other options is correct
- ☐ d. A specific classification head appended to the model
- ☐ e. The vocabulary embeddings of the model corresponding to natural language tokens
- ☐ f. Only the final projection layer of the model
- ☐ g. All transformer layers and the input embeddings of the model

Your answer is incorrect.

La risposta corretta è:

None of the other options is correct